Music sequencer

A **music sequencer** (or **audio sequencer**[1] or simply **sequencer**) is a device or application software that can record, edit, or play back music, by handling note and performance information in several forms, typically CV/Gate, MIDI,[2] or Open Sound Control, and possibly audio and automation data for digital audio workstations (DAWs) and plug-ins.

Overview

Modern sequencers

The advent of Musical Instrument Digital Interface (MIDI)[4] in the 1980s gave programmers the opportunity to design software that could more easily record and play back sequences of notes played or programmed by a musician. As the technology matured, sequencers gained more features, such as the ability to record multitrack audio. Sequencers used for audio recording are called digital audio workstations (DAWs).

Many modern sequencers can be used to control virtual instruments implemented as software plug-ins. This allows musicians to replace expensive and cumbersome standalone synthesizers with their software equivalents.

Today the term *sequencer* is often used to describe software. However, hardware sequencers still exist. Workstation keyboards have their own proprietary built-in MIDI sequencers. Drum machines and some older synthesizers have their own step sequencer built in. The market demand for standalone hardware MIDI sequencers has diminished greatly due to the greater feature set of their software counterparts.

Types of music sequencer

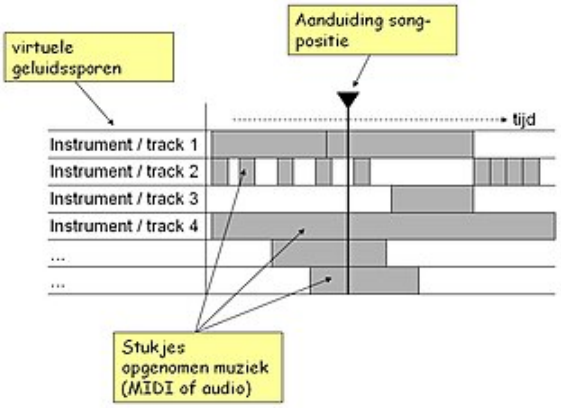
Music sequencers can be categorized by handling data types, such as:

- MIDI data for MIDI sequencers[5]
- CV/Gate data for analog sequencers[6][a]
- Automation data for mixing-automation in DAWs,[7][b] and software effect or instrument plug-ins for DAWs with sequencing features
- Audio data in audio sequencers[9] including DAWs, loop-based music software, etc.; or phrase samplers including grooveboxes, etc.

Also, a music sequencer can be categorized by its construction and supported modes.

1980s typical software sequencer platform, using Atari Mega ST computer

Today's typical software sequencer, supporting multitrack audio and plug-ins (Steinberg Cubase 6[3])



User interface on Steinberg Cubase 6, a digital audio workstation with an integrated software sequencer

Analog sequencer

Analog sequencers are typically implemented with analog electronics, and play the musical notes designated by a series of knobs or sliders for adjusting the note corresponding to each step in the sequence. It is designed for both composition and live performance; users can change the musical notes at any time without regard to recording mode. The time interval between each musical note (length of each step) may be independently adjustable. Typically, analog sequencers are used to generate repeated minimalistic phrases which may be reminiscent of Tangerine Dream, Giorgio Moroder or trance music.

An analog sequencer

Step sequencer (step recording mode)

On step sequencers, musical notes are rounded into steps of equal time intervals, and users can enter each musical note without exact timing; Instead, the timing and duration of each step can be designated in several different ways:

- On the drum machines: select a trigger timing from a row of *step-buttons*.
- On the bass machines: select a step note (or rest) from a *chromatic keypad*, then select a step duration (or tie) from a group of *length-buttons*, sequentially.
- On the several home keyboards: in addition to the real-time sequencer, a pair of *step trigger buttons* is provided; using it, notes on the pre-recorded sequence can be triggered in arbitrary timings for the timing dedicated recordings or performances.

In general, step mode, along with roughly quantized semi-realtime mode, is often supported on the drum machines, bass machines and several groove machines.

A *step rhythm sequencer* on the drum machine

A *step note sequencer* on the bass machine

Realtime sequencer (realtime recording mode)

Realtime sequencers record the musical notes in real-time as on audio recorders, and play back musical notes with designated tempo, quantizations, and pitch. For editing, often punch in/out features originating in tape recording workflows are provided. This mode is widely supported on software sequencers, DAWs, and built-in hardware sequencers.

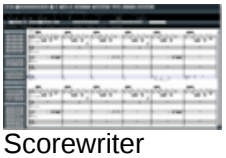
A realtime sequencer on the synthesizer

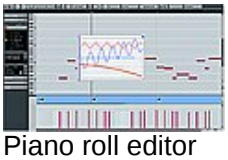
Software sequencer

A *software sequencer* is application software providing the functionality of a music sequencer, and often provided as one feature of the DAW or the integrated music authoring environments. The user may control the software sequencer either by using the graphical user interfaces or a specialized input devices, such as a MIDI controller.

Typical features on software sequencers

Numerical editor on music tracker


Scorewriter


Piano roll editor with strip chart

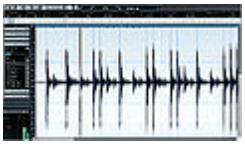
Audio and MIDI tracks on DAW

Automated, software studio environment including instruments and effect processors

Loop sequencer

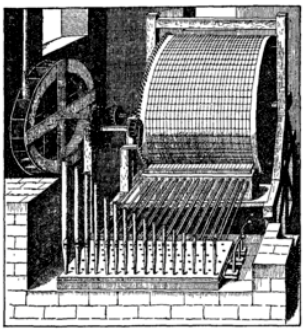
Audio sequencer

Alternative subsets of audio sequencers include:

	Digital audio workstation (DAW), hard disk recorder — a class of audio software or dedicated system primarily designed to record, edit, and play back digital audio, first appeared in the late 1970s and emerging since the 1990s. After the 1990s–2000s, several DAWs for music production were integrated with a music sequencer. *DAW integrated with MIDI sequencer* is often simply abbreviated as DAW, or sometimes referred to as *Audio and MIDI sequencer*,[10] *etc*. The term *audio sequencer* is a synonym for DAW.
	Loop-based music software — a class of music software for loop-based music compositions and remix, available since the late 1990s. Typical software included ACID Pro (1998), Ableton Live (2001) and GarageBand (2004). Some of these platforms are now considered DAWs due to expansions of feature sets or integrations. Looping's core feature, audio time stretching and pitch scaling allows users to control audio samples via MIDI in several aspects; user can designate pitches and durations independently on short music samples, as on MIDI notes.
	Tracker (music software) — a class of software music sequencer with embedded sample players, developed since the 1980s. Although it provides earlier "sequence of sampling sound" similar to grooveboxes and later loop-based music software, its design is slightly dated, and rarely referred as *audio sequencer*.
	Phrase sampler (or phrase sampling) — similar to above, musicians or remixers sometimes remixed or composed songs by sampling relatively long phrases or part of songs, and then rearranging these on grooveboxes or a combination of sampler (musical instrument) and sequencer. This technique is possibly referred as *audio sequencing*.
	Beat slicing — before the DAW became popular, several musicians sometimes derived various beats from limited drum sample loops by slicing beats and rearranging them on samplers. This technique, called *beat slicing*, was popularized with the introduction of *beat slicer* tool, especially the ReCycle released in 1992. Possibly it may be one origin of *audio sequencing*.

History

Early sequencers



Barrel with pins on a large stationary barrel organ

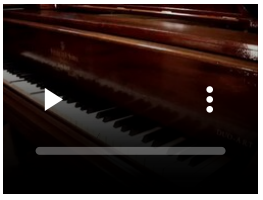
Music roll on barrel organ

The early music sequencers were sound-producing devices such as automatic musical instruments, music boxes, mechanical organs, player pianos, and Orchestrions. Player pianos, for example, had much in common with contemporary sequencers. Composers or arrangers transmitted music to piano rolls which were subsequently edited by technicians who prepared the rolls for mass duplication. Eventually consumers were able to purchase these rolls and play them back on their own player pianos.

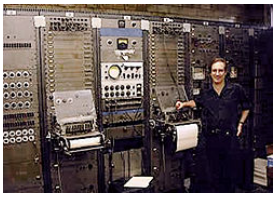
The origin of automatic musical instruments seems remarkably old. As early as the 9th century, the Persian (Iranian) Banū Mūsā brothers invented a hydropowered organ using exchangeable cylinders with pins,[11] and also an automatic flute-playing machine using steam power,[12][13] as described in their *Book of Ingenious Devices*. The Banu Musa brothers' automatic flute player was the first programmable music sequencer device,[14] and the first example of repetitive music technology, powered by hydraulics.[15]

In 1206, Al-Jazari, an Arab engineer, invented programmable musical automata,[16] a "robot band" which performed "more than fifty facial and body actions during each musical selection."[17] It was notably the first programmable drum machine. Among the four automaton musicians were two drummers. It was a drum machine where pegs (cams) bump into little levers that operated the percussion. The drummers could be made to play different rhythms and different drum patterns if the pegs were moved around.[18]

In the 14th century, rotating cylinders with pins were used to play a carillon (steam organ) in Flanders, and at least in the 15th century, barrel organs were seen in the Netherlands.[19]



Player piano (1920) controlled by piano roll

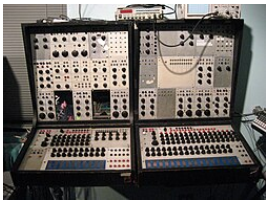


RCA Mark II (1957), controlled via wide punched-paper roll

In the late-18th or early-19th century, with technological advances of the Industrial Revolution various automatic musical instruments were invented. Some examples: music boxes, barrel organs and barrel pianos consisting of a barrel or cylinder with pins or a flat metal disc with punched holes; or mechanical organs, player pianos and orchestrions using book music / music rolls (piano rolls) with punched holes, etc. These instruments were disseminated widely as popular entertainment devices prior to the inventions of phonographs, radios, and sound films which eventually eclipsed all such home music production devices. Of them all, punched-paper-tape media had been used until the mid-20th century. The earliest programmable music synthesizers including the RCA Mark II Sound Synthesizer in 1957, and the Siemens Synthesizer in 1959, were also controlled via punch tapes similar to piano rolls.[20][21][22]

Additional inventions grew out of sound film audio technology. The drawn sound technique which appeared in the late 1920s, is notable as a precursor of today's intuitive graphical user interfaces. In this technique, notes and various sound parameters are triggered by hand-drawn black ink waveforms directly upon the film substrate, hence they resemble piano rolls (or the 'strip charts' of the modern sequencers/DAWs). Drawn soundtrack was often used in early experimental electronic music, including the Variophone developed by Yevgeny Sholpo in 1930, and the Oramics designed by Daphne Oram in 1957, and so forth.

Analog sequencers



Early commercially available analog sequencers (bottom) on Buchla 100 (1964/1966)[23]

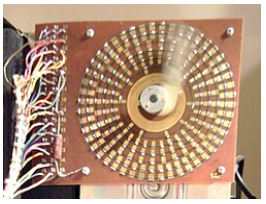
Moog sequencer module (top left, probably added after 1968) on Moog Modular (1964)

During the 1940s–1960s, Raymond Scott, an American composer of electronic music, invented various kind of music sequencers for his electric compositions. The "Wall of Sound", once covered on the wall of his studio in New York during the 1940s–1950s, was an electro-mechanical sequencer to produce rhythmic patterns, consisting of stepping relays (used on dial pulse telephone exchange), solenoids, control switches, and tone circuits with 16 individual oscillators.[24] Later, Robert Moog would explain it in such terms as "the whole room would go 'clack – clack – clack', and the sounds would come out all over the place".[25] The Circle Machine, developed in 1959, had incandescent bulbs each with its own rheostat, arranged in a ring, and a rotating arm with photocell scanning over the ring, to generate an arbitrary waveform. Also, the rotating speed of the arm was controlled via the brightness of lights, and as a result, arbitrary rhythms were generated.[26] The first electronic sequencer was invented by Raymond Scott, using thyratrons and relays.[27]

Clavivox, developed since 1952, was a kind of keyboard synthesizer with sequencer.[28] On its prototype, a theremin manufactured by young Robert Moog was utilized to enable portamento over 3-octave range, and on later version, it was replaced by a pair of photographic film and photocell for controlling the pitch by voltage.[25]

In 1968, Ralph Lundsten and Leo Nilsson had a polyphonic synthesizer with sequencer called Andromatic built for them by Erkki Kurenniemi.[29]

Step sequencers



Electro-mechanical disc sequencer on early drum machine (1959)

Eko ComputeRhythm (1972),[30][31] one of the earliest programmable drum machines

Firstman SQ-01 (1980),[32] one of the earliest step bass machines

The *step sequencer*s played rigid patterns of notes using a grid of (usually) 16 buttons, or steps, each step being 1/16 of a measure. These patterns of notes were then chained together to form longer compositions. Sequencers of this kind are still in use, mostly built into drum machines and grooveboxes. They are monophonic by nature, although some

are multi-timbral, meaning that they can control several different sounds but only play one note on each of those sounds.

Early computers

CSIRAC played the earliest computer music in 1951.

On the other hand, software sequencers were continuously utilized since the 1950s in the context of computer music, including computer-*played* music (software sequencer), computer-*composed* music (music synthesis), and computer *sound generation* (sound synthesis). In June 1951, the first computer music *Colonel Bogey* was played on CSIRAC, Australia's first digital computer.[33][34] In 1956, Lejaren Hiller at the University of Illinois at Urbana–Champaign wrote one of the earliest programs for computer music composition on ILLIAC, and collaborated on the first piece, *Illiac Suite for String Quartet*, with Leonard Issaction.[35] In 1957 Max Mathews at Bell Labs wrote MUSIC, the first widely used program for sound generation, and a 17-second composition was performed by the IBM 704 computer. Subsequently, computer music was mainly researched on the expensive mainframe computers in computer centers, until the 1970s when minicomputers and then microcomputers became available in this field.

In Japan

In Japan, experiments in computer music date back to 1962, when Keio University professor Sekine and Toshiba engineer Hayashi experimented with the TOSBAC computer. This resulted in a piece entitled *TOSBAC Suite*.[36]

Early computer music hardware

DDP-24 S Block (expansion card rack unit) that is assumed the A/D converters used for GROOVE (1970) by Max Mathews

In 1965,[37] Max Mathews and L. Rosler developed Graphic 1, an interactive graphical sound system (that implies sequencer) on which one could draw figures using a light-pen that would be converted into sound, simplifying the process of composing computer-generated music.[38][39] It used PDP-5 minicomputer for data input, and IBM 7094 mainframe computer for rendering sound.

Also in 1970, Mathews and F. R. Moore developed the GROOVE (Generated Real-time Output Operations on Voltage-controlled Equipment) system,[40] a first fully developed music synthesis system for interactive composition (that implies sequencer) and realtime performance, using 3C/Honeywell DDP-24[41] (or DDP-224[42]) minicomputers. It used a CRT display to simplify the management of music synthesis in realtime, 12-bit D/A converter for realtime sound playback, an interface for CV/gate analog devices, and even several controllers including a musical keyboard, knobs, and rotating joysticks to capture realtime performance.[38][42][39]

Digital sequencers

EMS Sequencer 256 (1971), branched from Synthi 100

In 1971, Electronic Music Studios (EMS) released one of the first digital sequencer products as a module of Synthi 100, and its derivation, Synthi Sequencer series.[43][44] After then, Oberheim released the DS-2 Digital Sequencer in 1974,[45] and Sequential Circuits released Model 800 in 1977 [46]

In Japan

Roland MC-8 MicroComposer (1977)

In 1977, Roland Corporation released the MC-8 MicroComposer, also called *computer music composer* by Roland. It was an early stand-alone, microprocessor-based, digital CV/gate sequencer,[47][48] and an early polyphonic sequencer.[49][50] It equipped a keypad to enter notes as numeric codes, 16 KB of RAM for a maximum of 5200 notes (large for the time), and a polyphony function which allocated multiple pitch CVs to a single Gate.[51] It was capable of eight-channel polyphony, allowing the creation of polyrhythmic sequences.[52][47][48] The MC-8 had a significant impact on popular electronic music, with the MC-8 and its descendants (such as the Roland MC-4 Microcomposer) impacting popular electronic music production in the 1970s and 1980s more than any other family of sequencers.[52] The MC-8's earliest known users were Yellow Magic Orchestra in 1978.[53]

Music workstations

Synclavier I (1977)

Fairlight CMI (1979) supporting MCL (sequence

In 1975, New England Digital (NED) released ABLE computer (microcomputer)[54] as a dedicated data processing unit for Dartmouth Digital Synthesizer (1973), and based on it, later Synclavier series were developed.

The Synclavier I, released in September 1977,[55] was one of the earliest digital music workstation product with multitrack sequencer. Synclavier series evolved throughout the late-1970s to the mid-1980s, and they also established integration of digital-audio and music-sequencer, on their Direct-to-Disk option in 1984, and later Tapeless Studio system.

Page R on Fairlight

In 1982, renewed the Fairlight CMI Series II and added new sequencer software Page R, which combined step sequencing with sample playback.[56]

While there were earlier microprocessor-based sequencers for digital polyphonic synthesizers,[c] their early products tended to prefer the newer internal digital buses than the old-style analogue CV/gate interface once used on their prototype system. Then in the early-1980s, they also re-recognized the needs of CV/gate interface, and supported it along with MIDI as options.

In Japan

Yamaha's GS-1, their first FM digital synthesizer, was released in 1980.[57]

MIDI sequencers

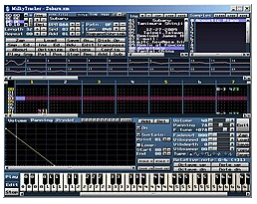
In June 1981, Roland Corporation founder Ikutaro Kakehashi proposed the concept of standardization between different manufacturers' instruments as well as computers, to Oberheim Electronics founder Tom Oberheim and Sequential Circuits president Dave Smith. In October 1981, Kakehashi, Oberheim and Smith discussed the concept with representatives from Yamaha, Korg and Kawai.[58] In 1983, the MIDI standard was unveiled by Kakehashi and Smith.[59][60] The first MIDI sequencer was the Roland MSQ-700, released in 1983.[61]

It was not until the advent of MIDI that general-purpose computers started to play a role as sequencers. Following the widespread adoption of MIDI, computer-based MIDI sequencers were developed. MIDI-to-CV/gate converters were then used to enable analogue synthesizers to be controlled by a MIDI sequencer.[48] Since its introduction, MIDI has remained the musical instrument industry standard interface through to the present day.[62]

Personal computers

In 1987, software sequencers called trackers were developed to realize the low-cost integration of sampling sound and interactive digital sequencer as seen on Fairlight CMI II Page R. They became popular in the 1980s and 1990s as simple sequencers for creating computer game music, and remain popular in the demoscene and chiptune music.

Moog Song Producer (1983)
MIDI & CV/Gate interface on SynAmp



Tracker software (developed since 1987)

Modern computer digital audio software after the 2000s, such as Ableton Live, incorporates aspects of sequencers among many other features.

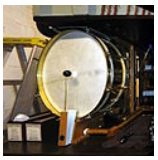
In Japan

In 1978, Japanese personal computers such as the Hitachi Basic Master equipped the low-bit D/A converter to generate sound which can be sequenced using Music Macro Language (MML).[63] This was used to produce chiptune video game music.[36]

It was not until the advent of MIDI, introduced to the public in 1983, that general-purpose computers really started to play a role as software sequencers.[48] NEC's personal computers, the PC-88 and PC-98, added support for MIDI sequencing with MML programming in 1982.[36] In 1983, Yamaha modules for the MSX featured music production capabilities,[64][65] real-time FM synthesis with sequencing, MIDI sequencing,[66][65] and a graphical user interface for the software sequencer.[67][65] Also in 1983, Roland Corporation's CMU-800 sound module introduced music synthesis and sequencing to the PC, Apple II,[68] and Commodore 64.[69]

The spread of MIDI on personal computers was facilitated by Roland's MPU-401, released in 1984. It was the first MIDI-equipped PC sound card, capable of MIDI sound processing[70] and sequencing.[71][72] After Roland sold MPU sound chips to other sound card manufacturers,[70] it established a universal standard MIDI-to-PC interface.[73] Following the widespread adoption of MIDI, computer-based MIDI software sequencers were developed.[48]

Visual timeline of rhythm sequencers



Mechanical (pre-20th century)

Rhythmicon (1930)

Drum machine (1959–)

Transistoriz machine (19

See also

- Combination action § Sequencers (for organs)
- Groovebox
- List of music sequencers
- List of music software
- Music tracker
- Music workstation

Notes

a. For a sequencer using a light source, see Circle Machine on #Analog sequencers and Raymond Scott#Electronics and research. and possibly others (via CV/Gate interfaces)

b. Automation parameters in DAWs are often interoperable with MIDI messages, i.e. Control Changes (CC) or System Exclusive (SysEx); in that case, it can be controlled in real-time via pre-assigned MIDI messages generated by MIDI controllers or MIDI sequencers, *etc*. And even more so, in several DAWs, automation parameters are explicitly recorded as MIDI messages on their embedded MIDI sequencers.[8]

c. In 1974–1975, Australian computer music engineer Tony Furse developed the MC6800-based Qasar M8 with a software sequencer MUSEQ 8, with a minimum price of $8,000. In 1976, it was licensed to Fairlight Instruments Pty Ltd., and eventually Fairlight CMI was released in 1979 (for details, see Fairlight CMI).
Also in 1975, New England Digital released original microprocessor-based ABLE computer (utilizing mini-computer architecture) as a future migration target of Dartmouth Digital Synthesizer. Their commercial version of digital synthesizer, Synclavier I was first shipped in 1977 (for details, see Synclavier).

References

1. Pejrolo, Andrea (2011). "1.7.1 The Primary Goals You Want to Achieve with Your Audio Sequencer" (https://books.google.com/books?id=k-vFPYjwpW0C&q=Music%20sequencer%20analog%20step%20digital%20realtime&pg=PR5). *Creative Sequencing Techniques for Music Production: A Practical Guide to Pro Tools, Logic, Digital Performer, and Cubase*. Taylor & Francis. p. 48. ISBN 978-0-240-52216-6.

2. Margaret Rouse (April 2005). "sequencer" (https://web.archive.org/web/20150627101306/http://whatis.techtarget.com/definition/sequencer). *WhatIs.com*. TechTarget. Archived from the original (https://whatis.techtarget.com/definition/sequencer) on 2015-06-27. "In digital audio recording, a sequencer is a program in a computer or stand-alone keyboard unit that puts together a sound sequence from a series (or sequence) of Musical Instrument Digital Interface (MIDI) events (operations). The MIDI sequencer allows the user to record and edit a musical performance without using an audio-based input source. ..."
3. "Cubase 6 screenshot licensed under CC-BY-SA-3.0" (https://www.steinberg.net/en/landing_pages/c6_creative_commons). Steinberg Media Technologies GmbH. Archived (https://web.archive.org/web/20111109011956/http://www.steinberg.net/en/landing_pages/c6_creative_commons) from the original on 2011-11-09.
4. Swift, Andrew. (May 1997), "A brief Introduction to MIDI" (https://web.archive.org/web/20120830211425/http://www.doc.ic.ac.uk/~nd/surprise_97/journal/vol1/aps2/), *SURPRISE*, Imperial College of Science Technology and Medicine, archived from the original (http://www.doc.ic.ac.uk/~nd/surprise_97/journal/vol1/aps2/) on 30 August 2012, retrieved 22 August 2012
5. Rothstein, Joseph (1995). *MIDI: A Comprehensive Introduction*. Computer Music and Digital Audio Series. Vol. 7. A-R Editions, Inc. pp. 77 (https://books.google.com/books?id=ajDaXh-qgDUC&pg=PA77), 122 (https://books.google.com/books?id=ajDaXh-qgDUC&pg=PA122). ISBN 978-0-89579-309-6.
6. Pinch, Trevor. J.; Trocco, Frank (2009). "Buchla's Box" (https://books.google.com/books?id=CoUs2SSvG4EC&pg=PT48). *Analog Days: The Invention and Impact of the Moog Synthesizer* (reprint ed.). Harvard University Press. pp. 55–56 (https://books.google.com/books?id=CoUs2SSvG4EC&dq=analog%20sequencer%20synthesizer&pg=PT55). ISBN 978-0-674-04216-2. "Subotnick suggested that using a light source to control sound might be promising. ... Later he [Buchla] turned this into an electro mechanical sequencer by introducing step relays and a dial. ... Buchla, like Moog, realized that voltage control ... But Buchla was after something different; ... Buchla was led to the electronic sequencer—a device that later was used to make much influential pop, rock, and dance music. A sequencer produces predetermined control voltages in a cycle or sequence and can endlessly recycle ..."
7. Price, Simon. "Using Mixer Automation In Reason – Reason Tips & Techniques" (https://www.soundonsound.com/sos/sep06/articles/reasontech_0906.htm). Technique: Reason Notes. *Sound on Sound*. No. September 2006. Archived (https://web.archive.org/web/20160310132431/http://www.soundonsound.com/sos/sep06/articles/reasontech_0906.htm) from the original on 2016-03-10. "Controller Data Vs. Automation / ... sequencer package such as Logic or Pro Tools, ... are akin to automation on professional hardware mixing consoles, ... This type of automation system is different to using MIDI Continuous Controller [Control Changes] (CC) data, ... In Reason, automation is MIDI Controller [Control Changes] data, but with some specialised tools for handling the data and playing it back. ...,
Recording Mixer Automation / As automation in Reason is MIDI CC data, it must be recorded on a sequencer track."
8. Price 2006
9. Pejrolo, Andrea (2011). "1.7.1 The Primary Goals You Want to Achieve with Your Audio Sequencer" (https://books.google.com/books?id=k-vFPYjwpW0C&q=Music%20sequencer%20analog%20step%20digital%20realtime&pg=PR5). *Creative Sequencing Techniques for Music Production: A Practical Guide to Pro Tools, Logic, Digital Performer, and Cubase*. Taylor & Francis. p. 48. ISBN 978-0-240-52216-6. (sub-section title contains the expression *Audio Sequencer*)
10. *MusE – The open source sequencer* (https://muse-sequencer.org), "MusE is a MIDI/Audio sequencer with recording and editing capabilities ..."
11. Fowler, Charles B. (October 1967). "The Museum of Music: A History of Mechanical Instruments". *Music Educators Journal*. **54** (2): 45–49. doi:10.2307/3391092 (https://doi.org/10.2307%2F3391092). JSTOR 3391092 (https://www.jstor.org/stable/3391092). S2CID 190524140 (https://api.semanticscholar.org/CorpusID:190524140).

12. Koetsier, Teun (2001). "On the prehistory of programmable machines: musical automata, looms, calculators". *Mechanism and Machine Theory*. **36** (5): 589–603. doi:10.1016/S0094-114X(01)00005-2 (https://doi.org/10.1016%2FS0094-114X%2801%2900005-2).
13. Banu Musa (1979). *The book of ingenious devices (Kitāb al-ḥiyal)*. Translated by Donald Routledge Hill. Springer. pp. 76–7. ISBN 9027708339.
14. Long, Jason; Murphy, Jim; Carnegie, Dale; Kapur, Ajay (12 July 2017). "Loudspeakers Optional: A history of non-loudspeaker-based electroacoustic music" (https://doi.org/10.1017%2FS1355771817000103). *Organised Sound*. **22** (2). Cambridge University Press: 195–205. doi:10.1017/S1355771817000103 (https://doi.org/10.1017%2FS1355771817000103).
15. Levaux, Christophe (12 July 2017). "The Forgotten History of Repetitive Audio Technologies" (https://doi.org/10.1017%2FS1355771817000097). *Organised Sound*. **22** (2). Cambridge University Press: 187–194. doi:10.1017/S1355771817000097 (https://doi.org/10.1017%2FS1355771817000097). hdl:2013/ULB-DIPOT:oai:dipot.ulb.ac.be:2013/385349 (https://hdl.handle.net/2013%2FULB-DIPOT%3Aoai%3Adipot.ulb.ac.be%3A2013%2F385349).
16. Fowler, Charles B. (October 1967). "The Museum of Music: A History of Mechanical Instruments". *Music Educators Journal*. **54** (2): 45–49. doi:10.2307/3391092 (https://doi.org/10.2307%2F3391092). JSTOR 3391092 (https://www.jstor.org/stable/3391092). S2CID 190524140 (https://api.semanticscholar.org/CorpusID:190524140).
17. Fowler, Charles B. (October 1967), "The Museum of Music: A History of Mechanical Instruments", *Music Educators Journal*, **54** (2), MENC_ The National Association for Music Education: 45–49, doi:10.2307/3391092 (https://doi.org/10.2307%2F3391092), JSTOR 3391092 (https://www.jstor.org/stable/3391092), S2CID 190524140 (https://api.semanticscholar.org/CorpusID:190524140)
18. Noel Sharkey, A 13th Century Programmable Robot (Archive) (https://web.archive.org/web/20070629182810/http://www.shef.ac.uk/marcoms/eview/articles58/robot.html), University of Sheffield.
19. Schlesinger, Kathleen (1911). "Barrel-organ" (https://en.wikisource.org/wiki/1911_Encyclop%C3%A6dia_Britannica/Barrel-organ). In Chisholm, Hugh (ed.). *Encyclopædia Britannica*. Vol. 3 (11th ed.). Cambridge University Press. pp. 432–434.
20. "The RCA Synthesiser" (https://120years.net/machines/rca/). *120 Years of Electronic Music (120years.net)*. 2014-02-11. Archived (https://archive.wikiwix.com/cache/20111026223002/http://120years.net/machines/rca/) from the original on 2011-10-26.—(PDF version (https://www.mathieubosi.com/zikprojects/120YearsOfElectronicMusic.pdf) Archived (https://web.archive.org/web/20120402113013/http://www.mathieubosi.com/zikprojects/120YearsOfElectronicMusic.pdf) 2012-04-02 at the Wayback Machine is available)
21. "Das Siemens-Studio für elektronische Musik von Alexander Schaaf und Helmut Klein" (https://web.archive.org/web/20130930175905/http://www.deutsches-museum.de/sammlungen/ausgewaehlte-objekte/meisterwerke-vi/siemens-studio) (in German). Deutsches Museum. Archived from the original (https://www.deutsches-museum.de/sammlungen/ausgewaehlte-objekte/meisterwerke-vi/siemens-studio/) on 2013-09-30.
22. Holmes, Thom (2012). "Early Synthesizers and Experimenters" (https://books.google.com/books?id=aT5nAQAAQBAJ&pg=PA175). *Electronic and Experimental Music: Technology, Music, and Culture* (4th ed.). Routledge. pp. 190 (https://archive.org/details/electronicexperi00holm/page/190)–192 (https://archive.org/details/electronicexperi00holm/page/192). ISBN 978-1-136-46895-7. See also excerpt from pp. 157-160 in Chapter 6 (http://static1.squarespace.com/static/50e79ec7e4b07dba60068e4d/t/5157 04dce4b05239ba84e64c/1364657372465/Holmes4.pdf) of *Early Synthesizers and Experimenters*.
23. Holmes, Thom (2008). *Electronic and experimental music: technology, music, and culture* (3rd ed.). Routledge. p. 222 (https://books.google.com/books?id=Q0uTAgAAQBAJ&q=Buchla+100+sequencer&pg=PA222). ISBN 978-0-415-95781-6. "Moog admired Buchla's work, recently stating that Buchla designed a system not only for "making new sounds but [for] making textures out of these sounds by specifying when these sounds could change and how regular those change would be.""

24. "Wall of Sound (sequencer)" (https://web.archive.org/web/20111113023228/http://raymondscott.com/1946.htm). *RaymondScott.com*. Archived from the original (https://raymondscott.com/1946.htm) on 2011-11-13.
25. Moog, Robert. "Memories of Raymond Scott" (https://web.archive.org/web/20111106035451/http://raymondscott.com/moog.html). *RaymondScott.com*. Archived from the original (https://raymondscott.com/moog.html) on 2011-11-06.
26. "Circle Machine" (https://web.archive.org/web/20110927232227/http://raymondscott.com/circle.html). *RaymondScott.com*. Archived from the original (https://raymondscott.com/circle.html) on 2011-09-27.—includes 2 sound files: Raymond Scott's demonstration, and commercial soundtrack for new batteries of Ford Motors.
27. Raymond Scott Artifacts (https://www.raymondscott.net/docs/RS-Artifacts.pdf), p. 13
28. Miller, Paul D. (2008-03-14). *Sound Unbound: Sampling Digital Music and Culture* (https://books.google.com/books?id=9vvapHTNuqwC&dq=%22Clavivox%22+synthesizer+%22sequencer%22&pg=PA182). MIT Press. ISBN 978-0-262-26646-8.
29. Städje, Jörgen (2012-10-06). "Andromatic, den automatiska andromedaren" (https://www.idg.se/2.1085/1.445306/andromatic-den-automatiska-andromedaren). International Data Group (IDG). Archived (https://web.archive.org/web/20121007224640/http://www.idg.se/2.1085/1.445306/andromatic-den-automatiska-andromedaren) from the original on 2012-10-07.
30. "EKO Computerhythm (1972)" (https://www.jarrography.free.fr/details_equipement_audio.php?id_equip=117). *Jarrography – The ultimate Jean Michel Jarre discography*. Archived (https://web.archive.org/web/20120503190624/http://www.jarrography.free.fr/details_equipement_audio.php?id_equip=117) from the original on 2012-05-03.
31. "EKO Computerhythm" (https://www.synthmaster.de/ekodrum.htm). *SynthMaster.de*. Archived (https://web.archive.org/web/20160304083316/http://www.synthmaster.de/ekodrum.htm) from the original on 2016-03-04.
32. "Multivox International" (https://web.archive.org/web/20030420170643/http://www.synrise.de/docs/types/f/firstman.htm). *SYNRISE* (in German). Archived from the original (https://www.synrise.de/docs/types/f/firstman.htm) on 2003-04-20.
33. "CSIRAC: Australia's first computer" (https://www.csiro.au/science/ps4f.html). Australia: Commonwealth Scientific and Industrial Research Organisation (CSIRO). Archived (https://web.archive.org/web/20071116112251/http://www.csiro.au/science/ps4f.html) from the original on 2007-11-16. Retrieved 2007-12-21.
34. Fildes, Jonathan (2008-06-17). " 'Oldest' computer music unveiled" (https://news.bbc.co.uk/1/hi/technology/7458479.stm). *BBC News Online*. Archived (https://web.archive.org/web/20090111225358/http://news.bbc.co.uk/1/hi/technology/7458479.stm) from the original on 2009-01-11. Retrieved 2008-06-18.—another oldest known recording of computer realized music played by the Ferranti Mark 1, captured by BBC in Autumn, 1951; the songs *Baa Baa Black Sheep* and *In the Mood*.
35. Hiller, Lejaren (Winter 1981). "Composing with Computer: A Progress Report". *Computer Music Journal*. **5** (4): 7–21. doi:10.2307/3679501 (https://doi.org/10.2307%2F3679501). JSTOR 3679501 (https://www.jstor.org/stable/3679501).
also available in Curtis Roads, ed. (1992-10-08). *The Music Machine: Selected Readings from Computer Music Journal*. MIT Press (1989/1992). pp. 75 (https://books.google.com/books?id=bqKfS3qQjMQC&q=Mathews&pg=PA75). ISBN 978-0-262-68078-3.
36. Shimazu, Takehito (1994). "The History of Electronic and Computer Music in Japan: Significant Composers and Their Works". *Leonardo Music Journal*. **4**: 102–106. doi:10.2307/1513190 (https://doi.org/10.2307%2F1513190). JSTOR 1513190 (https://www.jstor.org/stable/1513190). S2CID 193084745 (https://api.semanticscholar.org/CorpusID:193084745).
37. Ninke, William (1965), "Graphic 1: A Remote Graphical Display Console System", *Proceedings of Fall Joint Computer Conference*, vol. 27

38. Holmes, Thom (2008). "Digital Synthesis and Computer Music". *Electronic and experimental music: technology, music, and culture*. Taylor & Francis. pp. 254 (https://books.google.com/books?id=hCthQ-bec-QC&q=Mathews+MUSIC+GROOVE+%22Graphic+1%22&pg=PA254). ISBN 978-0-415-95781-6.
39. Roads, Curtis (Winter 1980). "Interview with Max Mathews". *Computer Music Journal*. **4** (4). in Curtis Roads, ed. (1992-10-08). *The Music Machine: Selected Readings from Computer Music Journal*. MIT Press (1989/1992). pp. 5 (https://books.google.com/books?id=bqKfS3qQjMQC&q=Mathews&pg=PA5). ISBN 978-0-262-68078-3.
40. Max V., Mathews; F.R., Moore (1970). "GROOVE—a program to compose, store, and edit functions of time". *Communications of the ACM*. **13** (12).
41. Nyssim Lefford; Eric D. Scheirer & Barry L. Vercoe. "An Interview with Barry Vercoe" (https://www.media.mit.edu/events/EMS/bv-interview.html). *Experimental Music Studio 25*. Machine Listening Group, MIT Media Laboratory. Archived (https://web.archive.org/web/20120331173352/http://www.media.mit.edu/events/EMS/bv-interview.html) from the original on 2012-03-31.
42. Bogdanov, Vladimir (2001). *All music guide to electronica: the definitive guide to electronic music*. Backbeat Books. pp. 320 (https://books.google.com/books?id=GJNXLSBlL7IC&q=GROOVE+Generated+Real-time+Output+Operations+on+Voltage-controlled+Equipment&pg=PT271). ISBN 978-0-87930-628-1.
43. Hinton, Graham (2001). "Synthi 100 (1971, formerly Digitana, aka the Delaware)" (https://www.ems-synthi.demon.co.uk/emsprods.html#synthi100). Electronic Music Studios (Cornwall). Archived (https://web.archive.org/web/20131031040324/http://www.ems-synthi.demon.co.uk/emsprods.html#synthi100) from the original on 2013-10-31.
44. Hinton, Graham (2001). "Synthi Sequencer 256 (1971, formerly Synthi Moog Sequencer)" (https://www.ems-synthi.demon.co.uk/emsprods.html#seq256). Electronic Music Studios (Cornwall). Archived (https://web.archive.org/web/20131031040324/http://www.ems-synthi.demon.co.uk/emsprods.html#seq256) from the original on 2013-10-31.
45. J.Michmerhuizen; Thomas E. Oberheim (June 1974). *DS-2 Digital Sequencer Instruction and Service Manual* (https://web.archive.org/web/20111218194017/http://www.cem3374.com/docs/Manuals/Oberheim/DS2_O%26SM.pdf) (PDF). Archived from the original (https://www.cem3374.com/docs/Manuals/Oberheim/DS2_O&SM.pdf) (PDF) on 2011-12-18. Retrieved 2017-12-06.
46. "Model 800 Sequencer" (https://web.archive.org/web/20111011114838/http://synthmuseum.com/sequ/seqseq80001.html). SynthMuseum.com. Archived from the original on 2011-10-11.
47. Russ, Martin (2008). *Sound Synthesis and Sampling* (https://books.google.com/books?id=_D2cTt5DPmEC&pg=PA346). Focal Press. p. 346. ISBN 978-0240521053. Retrieved 21 June 2011.
48. Russ, Martin (2012). *Sound Synthesis and Sampling* (https://books.google.com/books?id=X9h5AgAAQBAJ&pg=PA192). CRC Press. p. 192. ISBN 978-1136122149. Retrieved 26 April 2017.
49. Paul Théberge (1997), *Any Sound You Can Imagine: Making Music/Consuming Technology*, page 223 (https://books.google.com/books?id=asBnYmKKz6kC&pg=PA223), Wesleyan University Press
50. Herbert A. Deutsch (1985), *Synthesis: an introduction to the history, theory & practice of electronic music* (https://books.google.com/books?id=tjEJAQAAMAAJ), page 96, Alfred Music
51. Reid, Gordon. "The History Of Roland Part 1: 1930–1978" (https://www.soundonsound.com/sos/nov04/articles/roland.htm). *Sound on Sound* (Nov 2004). Archived (https://web.archive.org/web/20110629214447/http://www.soundonsound.com/sos/nov04/articles/roland.htm) from the original on 2011-06-29. Retrieved 2011-06-19.

52. Chris Carter, ROLAND MC8 MICROCOMPOSER (https://www.chriscarter.co.uk/content/sos/roland_mc8.html) Archived (https://web.archive.org/web/20170420095502/http://www.chriscarter.co.uk/content/sos/roland_mc8.html) 2017-04-20 at the Wayback Machine, *Sound on Sound*, vol.12, no.5, March 1997
53. *Yellow Magic Orchestra—Yellow Magic Orchestra (https://www.discogs.com/release/453067)* at Discogs
54. "Synclavier Early History" (http://www.500sound.com/synclavierhistory.html). Synclavier European Services. Archived (https://web.archive.org/web/20161114050731/http://www.500sound.com/synclavierhistory.html) from the original on 2016-11-14.
55. Chadabe, Joel (May 1, 2001). "The Electronic Century Part IV: The Seeds of the Future" (https://web.archive.org/web/20091002070017/http://emusician.com/tutorials/electronic_century4/). *Electronic Musician*. Archived from the original (https://emusician.com/tutorials/electronic_century4/) on October 2, 2009. "In September 1977, I bought the first Synclavier..."
56. "Fairlight – The Whole Story" (https://www.anerd.com/fairlight/fairlightstory.htm). *Audio Media*. No. January 1996. Archived (https://web.archive.org/web/20170504030342/http://www.anerd.com/fairlight/fairlightstory.htm) from the original on 2017-05-04. "*Fairlight launched the CMI Series II in 1982, which incorporated their now legendary Page R, the first serious music sequencer, which, according to Paine, "simply blew people away".*"
57. Roads, Curtis (1996). *The computer music tutorial* (https://books.google.com/books?id=nZ-TetwzVcIC&pg=PA226). MIT Press. p. 226. ISBN 0-262-68082-3. Retrieved 2011-06-05.
58. Chadabe, Joel (1 May 2000). "Part IV: The Seeds of the Future" (https://web.archive.org/web/20120928230435/http://www.emusician.com/gear/0769/the-electronic-century-part-iv-the-seeds-of-the-future/145415). *Electronic Musician*. **XVI** (5). Penton Media. Archived from the original (http://www.emusician.com/gear/0769/the-electronic-century-part-iv-the-seeds-of-the-future/145415) on 28 September 2012.
59. "Technical GRAMMY Award: Ikutaro Kakehashi And Dave Smith" (https://www.grammy.com/news/technical-grammy-award-ikutaro-kakehashi-and-dave-smith). 29 January 2013. Archived (https://web.archive.org/web/20160822073641/http://www.grammy.com/news/technical-grammy-award-ikutaro-kakehashi-and-dave-smith) from the original on 22 August 2016.
60. "Ikutaro Kakehashi, Dave Smith: Technical GRAMMY Award Acceptance" (https://www.grammy.com/videos/technical-grammy-award-recipients-ikutaro-kakehashi-and-dave-smith-at-special-merit-awards). 9 February 2013. Archived (https://web.archive.org/web/20141209022049/http://www.grammy.com/videos/technical-grammy-award-recipients-ikutaro-kakehashi-and-dave-smith-at-special-merit-awards) from the original on 9 December 2014.
61. "Roland - Company - History - History" (https://www.roland.com/ca/company/history/). Archived (https://web.archive.org/web/20170712075811/https://www.roland.com/ca/company/history/) from the original on 2017-07-12. Retrieved 2017-05-17.
62. The life and times of Ikutaro Kakehashi, the Roland pioneer modern music owes everything to (http://www.factmag.com/2017/04/02/ikutaro-kakehashi-life/) Archived (https://web.archive.org/web/20170403005512/http://www.factmag.com/2017/04/02/ikutaro-kakehashi-life/) 2017-04-03 at the Wayback Machine, *Fact*
63. Kunihiko, Nagai; Teruhiro, Takezawa; Kazuma, Yoshimura; KaTsutoshi, Tajima (April 1979). "Micro computer Basic Master (MB-6880)" (https://www.hitachihyoron.com/jp/pdf/1979/04/1979_04_26.pdf) (PDF) (in Japanese). Archived (https://web.archive.org/web/20170915071518/http://www.hitachihyoron.com/jp/pdf/1979/04/1979_04_26.pdf) (PDF) from the original on 2017-09-15. "***2.特長*** *... (4) スピーカーを内蔵しており、プログラムによる音楽の自動演奏が可能である。 /* ***表 I*** *「ベーシックマスター」の主な仕様一覧 ... 音楽発生機能: 5ビットD/A変換のスピーカー再生 /* ***4.3 音楽発生機能*** *...*"
Published on: "Special Features: Micro computer and its application" (https://www.hitachihyoron.com/jp/archive/1970s/1979/04.html). *Hitachi Hyoron* (April 1979). Japan: Hitachi. Archived (https://web.archive.org/web/20170915071500/http://www.hitachihyoron.com/jp/archive/1970s/1979/04.html) from the original on 15 September 2017. Retrieved 15 September 2017.

64. Martin Russ, *Sound Synthesis and Sampling*, page 84 (https://books.google.com/books?id=X9h5AgAAQBAJ&pg=PA84), CRC Press
65. David Ellis, Yamaha CX5M (http://www.muzines.co.uk/articles/yamaha-cx5m/1481) Archived (https://web.archive.org/web/20171026002632/http://www.muzines.co.uk/articles/yamaha-cx5m/1481) 2017-10-26 at the Wayback Machine, *Electronics & Music Maker*, October 1984
66. *Yamaha Music Computer CX5M Owner's Manual* (https://web.archive.org/web/20151022170650/http://download.yamaha.com/api/asset/file/?language=hu&site=hu.yamaha.com&asset_id=4605). Yamaha. Archived from the original (https://download.yamaha.com/api/asset/file/?language=hu&site=hu.yamaha.com&asset_id=4605) on 2015-10-22. Retrieved 2018-12-26.
67. Yamaha (5 May 1984). *Yamaha CX5M Music Computer Flyer (GB)* (https://archive.org/details/CX5MFlyer). Retrieved 5 May 2018 – via Internet Archive.
68. Roland CMU-800 (http://www.vintagesynth.com/roland/cmu800.php) Archived (https://web.archive.org/web/20170604041554/http://www.vintagesynth.com/roland/cmu800.php) 2017-06-04 at the Wayback Machine, Vintage Synth Explorer
69. Happy birthday MIDI 1.0: Slave to the rhythm (https://www.theregister.co.uk/Print/2013/08/26/part_two_midi_spec_1_is_30_happy_birthday_musical_instrument_digital_interface/) Archived (https://web.archive.org/web/20171026002557/http://www.theregister.co.uk/Print/2013/08/26/part_two_midi_spec_1_is_30_happy_birthday_musical_instrument_digital_interface/) 2017-10-26 at the Wayback Machine, *The Register*
70. MIDI INTERFACES FOR THE IBM PC (http://www.textfiles.com/music/midi-em.txt) Archived (https://web.archive.org/web/20151021050032/http://textfiles.com/music/midi-em.txt) 2015-10-21 at the Wayback Machine, *Electronic Musician*, September 1990
71. "Programming the MPU-401" (https://www.piclist.com/techref/io/serial/midi/mpu.html). *www.piclist.com*. Archived (https://web.archive.org/web/20170506080336/http://www.piclist.com/techref/io/serial/midi/mpu.html) from the original on 6 May 2017. Retrieved 5 May 2018.
72. MIDI PROCESSING UNIT MPU-401 TECHNICAL REFERENCE MANUAL (https://ftp.oldskool.org/pub/drivers/Roland/MPU-401%20technical%20reference%20manual.pdf), Roland Corporation
73. Peter Manning (2013), *Electronic and Computer Music* (https://books.google.com/books?id=ryet1i-8OlYC), page 319, Oxford University Press

Further reading

List of papers sharing a similar perspective with this Wikipedia article:

- Arar, Raphael; Kapur, Ajay (2013). "A History of Sequencers: Interfaces for Organizing Pattern-Based Music" (https://web.archive.org/web/20150402095540/http://smcnetwork.org/system/files/A%20HISTORY%20OF%20SEQUENCERS%20INTERFACES%20FOR%20ORGANIZING%20PATTERN-BASED%20MUSIC.pdf) (PDF). *Proceedings of the Sound and Music Computing Conference 2013 (SMC 2013), Stockholm, Sweden*. Archived from the original (https://smcnetwork.org/system/files/A%20HISTORY%20OF%20SEQUENCERS%20INTERFACES%20FOR%20ORGANIZING%20PATTERN-BASED%20MUSIC.pdf) (PDF) on 2015-04-02.
  Note: although this conference paper emphasized the "Ace Tone FR-1 Rhythm Ace", it is not the music sequencer nor the first drum machine product.

External links

- "History of electronic musical instruments and sequencers" (https://www.120years.net). *120 Years of Electronic Music (120years.net)*.

- "Early sequencer controllers" (https://www.vintagesynth.com/misc/sequencers.shtml). *Vintage Synth Explorer*.
- Richmond, Leigh (11 November 1974). "Computer hums its own music" (http://www.exoticsciences.com/ra_scrpbk/musicx.htm). *Evening Times*. Melbourne, FL. p. A1. (1974 newspaper article about digital sequencer)
- "German National Museum of Musical Automata in Bruchsal palace" (https://www.landesmuseum.de/website/Deutsch/Sammlungsausstellungen/Deutsches_Musikautomaten-Museum/Ausstellung.htm), *website of the Badisches Landesmuseum*

Retrieved from "https://en.wikipedia.org/w/index.php?title=Music_sequencer&oldid=1350566694"